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INTERVENTION, EVALUATION, AND POLICY STUDIES

A Randomized Trial of Teacher Development in Elementary Science: First-Year Achievement Effects

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Abstract: This article describes 1st-year experimental effects of a large-scale reform providing professional development to elementary school teachers to implement an extended, inquiry-oriented science curriculum. Known as "immersion teaching" because it "immerses" teachers and students in the full cycle of scientific inquiry, this approach developed through a partnership involving university-based science and mathematics content experts and educators and K-12 educators from the Los Angeles Unified School District. Multilevel analyses, which examined school-level effects of assignment to the professional development intervention, nested Grade 4 students and their science achievement outcomes within the 80 study schools. The analyses revealed a statistically significant negative 1st-year treatment effect of school-level assignment to the initiative on the key science achievement outcome. We also tested whether the treatment had differential effects for English language learners, schools with large proportions of English language learners, and students of new teachers. We found an interaction effect of the treatment by teacher experience level for the teachers who were the primary target of the intervention, with the treatment having positive effects for novice teachers (3 years of experience or less) but a larger, negative effect for veteran teachers. We explore analytically three sets of explanations for the unexpected negative main effect of treatment: potential statistical and design artifacts, possible misalignment between the assessments and content of the treatment, and practical issues related to implementation of the treatment.

Keywords: Randomized experiment, science intervention, teacher professional development, inquiry-based pedagogy

The central goal of this study was to test experimentally the impact on student achievement of a content-rich, inquiry-oriented, systemic intervention in

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teacher development for elementary school science in a large, urban school district. System-Wide Change is a broad-based approach to science teaching and learning that involves a partnership among university scientists, science educators, and K–12 practitioners; addresses preservice, in-service, and curricular development; and is supported by a comprehensive National Science Foundation Math and Science Partnership known as SCALE (System-Wide Change for All Learners and Educators). We set out to evaluate the achievement effects of the System-Wide Change elementary science component, which provides fourth- and fifth-grade teachers with professional development in summer institutes and ongoing coaching and mentoring in the use of extended, inquiry-based curriculum units for elementary science. The curriculum units, which are known as "immersion units," bring teachers and students through a full cycle of inquiry in core problems of scientific investigation.

This experiment contrasts (a) schools randomized to the teacher development treatment whose fourth- and fifth-grade science lead teachers were offered the opportunity to attend a summer institute and whose fourth- and fifth-grade teachers were targeted for coaching and mentoring to implement the immersion units, with (b) control schools whose fourth- and fifth-grade teachers were offered the immersion curriculum but not the associated professional development. This study estimates the causal impact of the teacher development activities on student learning of standards-based science. It takes place in the Los Angeles Unified School District (LAUSD), the largest district in the SCALE partnership and a context in which raising test scores in science is a great need and high priority. In this initial analysis of the achievement impacts of the intervention, we estimate school-level intention-to-treat effects of assignment for schools' fourth-grade teachers and students. In the 1st year of the study, the 2006–07 school year, the fourth-grade science lead teachers were targeted for the summer 2006 institutes offered by the SCALE team. Our analyses respond to the central research question: What is the impact of a content-rich, inquiry-based approach to teacher development on student achievement in fourth-grade science?

A NATIONAL NEED

There is a clear national need for improving elementary science teaching and learning, and the need is particularly great in urban districts that serve high proportions of disadvantaged students. A broad consensus has come to recognize that the preeminence of the United States as a source of technological ingenuity and economic growth rests substantially on the preparation of young persons to engage in fields requiring scientific literacy (e.g., American Association for the Advancement of Science, 1989). Yet recent national assessments of children's emerging knowledge and understanding of scientific concepts have yielded results that are worrisome. According to the National Assessment of Educational

Progress (NAEP), in 2000 only 29% of U.S. fourth graders scored at levels judged proficient in science for their grade, a figure that was unchanged from the previous assessment 4 years earlier (O'Sullivan, Lauko, Grigg, Qian, & Zhang, 2003). Unlike reading and mathematics trends, which may be responding to concerted improvement efforts, elementary science achievement trends are flat. Moreover, pervasive inequalities among students from different racial and economic groups have shown no signs of abating in recent years. In 2000, whereas 38% of White individuals scored proficient or above on NAEP fourth-grade science, the corresponding figures for Hispanics and African Americans were just 11% and 7%, respectively (O'Sullivan et al., 2003). Only 11% of students eligible for free or reduced-price lunch scored proficient, compared to 38% of those not eligible.

In California, students are not faring any better on science achievement tests. According to recent NAEP data, only 17% of fourth graders performed at a level deemed proficient or above compared to the national average of 29%. This places California near the bottom of state rankings in fourth-grade science scores, with only Mississippi scoring lower (http://nationsreportcard.gov/science_2005). These problems of low scores and wide achievement gaps are likely to become increasingly salient as science is tested under No Child Left Behind.

A FOCUS ON TEACHING FOR UNDERSTANDING

Why are students performing so poorly on these science standards, and what can our nation's educators and policymakers do to rectify this situation? Traditional elementary science instruction—in which science is conceived primarily as a body of content and a set of procedures and the goal of learning is to acquire the concepts and master the procedures—has failed to produce uniformly high levels of scientific literacy. Several major research endeavors point toward approaches aimed at fostering deeper understanding of core scientific ideas and an enhanced ability to reason scientifically (e.g., Gamoran et al., 2003; Glennan & Resnick, 2004; Romberg, Carpenter, & Dremock, 2005; Schunn, Millar, & Lauffer, & the SCALE Immersion Team, 2005). Common to all these endeavors is the notion that increased science achievement will require *learning with understanding*—that is, in-depth comprehension of powerful scientific ideas—and that learning with understanding requires *teaching for understanding*—that is, instruction guided by student reasoning about rigorous content in equitable classroom communities.

Prior efforts to encourage the adoption of rigorous, inquiry-based pedagogy among elementary science teachers in LAUSD were apparently unsuccessful. Although a small number of teachers were observed to demonstrate rigorous, inquiry-based science teaching, the vast majority of elementary teachers either taught science less than once per 3 days or used techniques that were not rigorous and/or inquiry-based (Hoffer & Cantrell, 2003; Kelly, 2004). Since these earlier studies, both the incentives and the resources for science teaching reform have changed. Federal and state accountability systems have added science as an area of emphasis, and LAUSD has responded to these accountability pressures by developing a plan of action for elementary science instruction, including immersion curricula for Grades 3 to 8. At least one module in each grade's science curriculum guide is an immersion unit that brings students into the full cycle of scientific inquiry in a content area related to state standards. For example, the immersion unit in Grade 4 focuses on the cycling of matter and the transfer of energy, elements of the Grade 4 state standards; the Grade 5 immersion unit centers on weather, which aligns well with the emphasis on extreme weather in the state standards for that grade.

The immersion units were developed and piloted through SCALE's unique collaboration of science and education faculty, school district instructional and science experts, and teachers. Immersion units are introduced with an immersion unit toolbox, which covers essential features of classroom inquiry, as drawn from the National Science Education Standards (National Research Council, 1996; Olson & Loucks-Horsley, 2000). Among these features are posing scientific questions, giving priority to evidence, connecting evidence-based explanations to scientific knowledge, and communicating and justifying explanations. The immersion units employ inquiry methods to address "big ideas"—core content and organizing principles of scientific understanding (Wilson & Berenthal, 2005).

Curricular materials provide a resource for improved practice, but they may not be enough to transform teaching (Gamoran et al., 2003). Indeed, recent research and pilot studies within LAUSD have suggested that a new curriculum alone may not suffice to change practice and raise test scores and, thus, the curricular units were not the focus of our randomized trial. Rather, our focus is on the district's effort to create *systemwide change in elementary science through teacher development*. In the context of increased state and federal accountability and higher district expectations reflected in the curriculum guides, the district has developed a support structure to build capacity for rigorous science teaching in its elementary teachers. These capacity-building efforts represent a significant departure from prior district efforts to encourage rigorous, inquiry-based practice, which relied on teacher volunteerism, giveaways, and word of mouth. In the past, a typical professional development session would be planned around a curricular unit or tool, such as a graphing calculator. The sessions would be advertised by informing local administrators who would encourage teachers to attend. To encourage participation, attendees were often promised materials for their classroom and/or units that they could immediately implement. These professional development sessions did not affect practice, in large part because of the lack of a coherent district plan for elementary science instruction. The sessions were frequently one-time events, lacking connection to previous or subsequent sessions and linked only by a general commitment to inquiry methods. Although not all of these features were undesirable, the net effect was to support pockets of excellence across the district, rather than broad changes in practice (Hoffer & Cantrell, 2003).

In this project, the immersion curricula are scaffolded by a system of teacher development in which teacher leaders from each school participate in an intensive summer institute and ongoing training and then share their learning with colleagues in their own schools. By supporting teachers in using the immersion units in their classrooms, district staff and their external partners aim to enhance teachers' abilities to teach science for understanding throughout the school year. In sum, the LAUSD is implementing a promising new model of teacher development that, when taken to scale, may promote systemwide change in science teaching and learning. The model has been piloted, setting the stage for this broader implementation that will permit testing effects on student achievement.

METHOD

Sample selection in the LAUSD began in winter 2006, with the preparation of field sites and background data collection occurring in spring 2006. A representative from the research team met with the LAUSD superintendent and the eight local superintendents who lead LAUSD's eight local districts. We argued that resource constraints coupled with a desire to implement the intervention with a high degree of fidelity made it necessary to implement System-Wide Change in yearly stages rather than all at once. These constraints are favorable for research because, most important, a staged implementation provides an opportunity to introduce a randomized element into the selection of participating schools. That is, because not all schools can be served at once, we can randomly select an initial set of treatment and control schools whose comparison will reveal the impact of the teacher development activities.

Our procedure for selecting schools reflected a compromise between the need to intervene in schools that are at least minimally ready to respond to the intervention, on one hand, and the desire to generalize within LAUSD as widely as possible, on the other. LAUSD's 420 elementary schools are distributed among the eight local districts. Each local district superintendent agreed to, in consultation with his or her principals, nominate 20 or more schools that were ready to participate in the intervention. This approach allowed the superintendents to hold back schools that were in great turmoil or undergoing leadership transitions but still offered a broad range of schools. A total of 191 schools were nominated by the eight local superintendents. These schools tended to have somewhat higher baseline science, math, and reading test scores than schools that were not nominated. In this respect, the nominated schools were, as we would expect based on the selection criteria, higher performing than the typical LAUSD elementary schools, The sample of 191 schools was too large for our study, so we chose a stratified (stratified by the eight local districts) random sample of 80 schools from the 191 that were nominated. These 80 schools were, thus, representative of the larger sample of 191 nominated schools. Using a block randomized design, with local districts as blocks, we randomly selected 5 schools from each local district for the intervention and 5 for the control group, for a total of 40 treatment and 40 control schools.

Sample

Thus, the sample for this analysis of science achievement outcomes consists of fourth-grade students nested within 80 schools. These 80 schools are evenly distributed over the eight local districts of the LAUSD. Five of these 80 schools are charter or magnet schools, but the rest are traditional. The majority of the sample schools contain kindergarten through fifth grade, with only 10 schools that have a slightly different grade distribution. Students attending sample schools are racially diverse, and many come from low-income families. Of students attending sample schools, about three fourths are Hispanic, nearly 9% are African American, approximately 9% are White, nearly 3% are Asian, 3% are Filipino, and less than 1% are Native American and Pacific Islanders. Approximately 81% qualify for free or reduced-price lunch, nearly 12% are served by special education services, and more than 42% are English language learners (ELLs). As the baseline descriptive statistics in Table 1 suggest, there were no statistically significant differences between control and treatment schools; they are comparable in terms of racial/ethnic composition, percentage eligible for free or reduced-price lunch, and science, reading, and math test scores.

Data

Two types of achievement indicators serve as dependent variables for the study. First, the district has developed periodic assessments aligned with the instructional guides and state standards, administered three times per year in Grades 4 to 8. Scores on these assessments in Grades 4 and 5 constitute one type of dependent variable. The three periodic assessments in Grades 4 and 5 follow the progression of the instructional guides by covering life science, earth science, and physical science (Scruggs, 2004). Each assessment consists of 20 multiple-choice questions and one constructed response. The actual sequencing of the instructional units—life science, earth science, and physical science—is determined by each teacher. Therefore, some teachers may decide to focus on life science as the first topic during the fall, whereas others may choose one of the other two areas as the initial focus. The administration of the periodic assessment follows the instructional unit and, thus, the sequencing of the administration of the three periodic assessments may also vary across classrooms and schools.

Table 1. Comparison of baseline characteristics of treatment ($N = 33$) and control schools ($N = 38$)

	Control	Treatment	F	p
Composite science achievement	0.569 (0.085)	0.577 (0.079)	0.163	0.690
Math achievement	355.960 (28.832)	359.712 (24.291)	0.346	0.559
Reading achievement	341.698 (21.532)	341.925 (19.759)	0.002	0.962
% free lunch	0.762 (0.263)	0.750 (0.255)	0.037	0.848
% English language learners	0.334 (0.246)	0.335 (0.185)	0.000	0.994
% African American	0.090 (0.159)	0.073 (0.082)	0.312	0.578
% White	0.122 (0.213)	0.104 (0.185)	0.152	0.698
% Native American	0.003 (0.008)	0.004 (0.009)	0.160	0.690
% Pacific Islander	0.007 (0.026)	0.005 (0.017)	0.161	0.690
% Asian	0.037 (0.061)	0.029 (0.054)	0.280	0.598
% Hispanic	0.690 (0.280)	0.763 (0.248)	1.324	0.254
% Filipino	0.051 (0.082)	0.022 (0.050)	2.982	0.089

Note. Standard deviations are in parentheses.

The LAUSD provided student scores for the science periodic assessments in the 2005–06 (pretreatment) and 2006–07 (Year 1, posttreatment) school years, as well as test administration dates for each student. Of interest, we observed a treatment effect on the ordering of the science instructional units. In the area targeted by the treatment, 61% of teachers, or 87 of 142, in treatment schools gave the life science test in the first 90 days of the school year, compared to 47%, or 61 of 130 teachers, in control schools. This difference was statistically significant, $\chi^2(1, N = 272) = 5.63$, $p < .05$. Only 20% (24/120) of teachers in treatment schools administered the earth science periodic assessment first compared to 30% (36/130) of teachers in control schools. This difference was also statistically significant, $\chi^2(1, N = 274) = 4.86$, $p < .05$. Finally, 24% of treatment and 19% of control teachers gave the physical science assessment in the first 90 days of the school year—this difference was not statistically significant, $\chi^2(1, N = 257) = 0.864$, $p = .22$. Because of this variation in sequencing the periodic assessments, we specified analytical models that adjusted for student-level differences in the timing of each test. Taking into account this variability associated with the treatment may explain differences in the outcomes because students taking the assessments during the latter half of the academic year may have somewhat better scores than students taking the tests during the first half of the school year because of general maturation and increased exposure to science instruction. Specifically, we constructed a student-specific indicator for each of the three periodic assessments to account for whether or not the particular instructional unit and test was administered during the first half of the academic year.

Our second outcome, which is not considered in this article, is the state standardized science test administered in Grade 5 as part of the overall California accountability system. We turn our focus to this outcome during the 2007–08 academic year as the fifth-grade teachers and students are targeted by the SCALE intervention. The assessment, known as the California Science Standards Test (CST-Science), is a norm-referenced test aligned with state standards (California State Department of Education, 2004) and is part of the state's reporting system under No Child Left Behind. Whereas the primary purpose of the periodic assessments is to provide feedback to teachers on student learning, the CST-Science also has an accountability purpose, and as such, it is a high-stakes test.

We anticipate that the teacher development activities will affect achievement on all these outcomes by influencing teachers' overall approach to science instruction. However, it seems possible the effects will be greatest for (or limited to) the topics addressed in the immersion units that will serve as instructional exemplars in the summer institutes. To address this possibility, we identify the periodic assessment that is related to the material covered in the immersion unit (life science in Grade 4, earth science in Grade 5) as the key outcomes.

School-level data consist of a pretest measure, a treatment status indicator, and indicators of the eight local districts, which also represent the randomization block. The pretest measure was the school mean on a composite measure of three periodic assessments administered during the 2005–06 school year. Although this variable actually measures the school average on all periodic assessments (life, physical, and earth science) for the previous year's cohort, it should still capture important observable school characteristics, which induce variation in the dependent variable. This pretest variable was constructed by averaging the student scores on each periodic assessment and then aggregating the student means to create a school mean.

Analytical Method

We applied a methodology that is relatively new to education: the *cluster randomized trial* (CRT) or, as referred to by Boruch et al. (2004), *place-based* *random assignment*. In education, CRTs randomize at a higher level of aggregation, such as the school or classroom, and collect data on individuals within the cluster, including teachers and students. CRTs are often the optimal design for studying the effects of school- and classroom-based interventions. They address practical problems related to program delivery and evaluation design, such as the potential difficulties of singling out and randomizing individual teachers within schools, or students within classrooms, to alternate treatments. Because all students and teachers at a site share the same experimental treatment status, CRTs also prevent crossovers of teachers or students from control to treatment, or vice versa. In addition, CRTs are well aligned with educational interventions that operate through coordinated, systemic initiatives delivered by organization-level elements acting in concert.

Estimation of treatment effects at the level of the cluster that was randomized is the appropriate method (Cornfield, 1978; Donner & Klar, 2000; Murray, 1998; Raudenbush, 1997). This method requires enough clusters to provide sufficient statistical power to estimates of treatment effects and statistical methods that are appropriate to clustered data. Raudenbush proposed the hierarchical linear model as an analytic strategy for CRTs. In our case, one may simultaneously model both student- and school-level sources of variability in each outcome. Specifically, we developed two-level hierarchical models of students nested within schools. The linear model for Level 1 of the analysis (students within schools) was written as

$$Y_{ij} = \beta_{0j} + \beta_{1j}(\text{TIMING})_{ij} + r_{ij},$$

which represents science achievement for student i in school j. The term r_{ij} is the Level 1 residual variance that remains unexplained after accounting for the timing of the particular periodic assessment (i.e., administered in the first half of the academic year or not).

Most important, the Level 2 model answered the question, Does the intervention affect the school-level achievement mean? Including the school-level aggregate prior-year composite periodic assessment, which was simply a mean of the three periodic assessments, improved the precision of the school-level treatment effect estimates. The fully specified Level 2 model was written as

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{PRETEST})_j + \gamma_{02}(\text{TREATMENT})_j + \gamma_{03\ldots09}(\text{DISTRICT})_j + u_{0j},$$
$$\beta_{1j} = \gamma_{10},$$

where the mean achievement intercept for school j, β_{0j}, is regressed on the school-level mean composite periodic assessment pretest, the treatment indicator, a vector of indicator variables representing the eight local districts, plus a residual, u_{0j}. Because we had no theoretical or practical reason to model the school-level outcome for the sequence dummy code, we treated it as fixed and predicted it by an intercept only.

These two-level models estimated school-level treatment effects of assignment to the teacher development program based on two primary groups of fourth-grade teachers and students: (a) all students and teachers from the study schools, and (b) only those students taught by the science lead teachers, who were primarily targeted for the professional development intervention. Because the role of science lead teacher in each treatment and control school was decided upon prior to randomization, these latter analyses are experimental contrasts just as are the schoolwide comparisons of all fourth-grade teachers and students.

RESULTS

We began by investigating the pattern of missing data and the extent to which data attrition may have affected the internal and external validity of the study. Of the 80 schools involved in this experiment, 78 administered the 2006–07 life science periodic assessment. In these schools, there were also students who did not take the life science periodic assessment. Although the district encourages teachers to administer the test, there is no penalty for not doing so. As a result, there were two whole schools, a number of whole classrooms of students, and individual students within classrooms who had missing outcome data. Overall, there were 3,091 of 4,016, or 76%, students in control schools who took the exam, and 3,555 of 4,503, or 79%, students in treatment schools who took the exam.

In addition to missing information on the outcome variable, three schools were missing pretest information because they did not administer any of the periodic exams in 2005–06. Three additional schools had two of the three periodic assessment scores used to construct the composite measure of science achievement; we averaged the two scores on the student level and then aggregated the averages to the school level, so that these schools could be included in the analyses. Finally, because the test timing dummy variable required a test date for the life, earth, and physical science periodic assessments, additional students dropped out of the analyses. In all, 6 schools were not included in the analysis of life science achievement. Of these, 1 is a control school and 5 are in the treatment group. Thus, these analyses were conducted with a sample of 6,385, which is 72.3% of all fourth-grade students from the 80 schools. The number of students missing in the analysis did not differ statistically between the treatment and control schools, $\chi^2(1, N = 8830) = 0.212$, $p = .33$.

Models with a life science outcome measure and an analytic sample restricted to students of science lead teachers included 1,526 students and 60 schools. Ten of these schools lost to attrition were in the treatment group, and 10 were in the control group. Despite this equivalence on the school level, there was differential attrition on the student level, with 777 of 1,084 (72%) students from the control group but only 749 of 1,166 (64%) of the students from the treatment group included in the analysis. This difference is statistically significant, $\chi^2(1, N = 2250) = 14.26$, $p < .01$. Because of this differential attrition, it is necessary to interpret findings from this sample with some caution. Given that the treatment group tested a smaller and more select group of students, this restricted sample may generate a positively biased estimate of the treatment impact.

Although the main outcome of interest is the life science periodic score, we also present some models for the purpose of elucidating how being in the treatment group may affect scores of the students of science lead teachers or may spill over to other content areas such as earth and physical science. Models with an earth science outcome measure and an unrestricted sample included 6,862 students and 74 schools, whereas those with a sample restricted to students of science lead teaches included 1,667 students and 63 schools. Finally, models with a physical science outcome measure and an unrestricted sample included 6,444 students and 71 schools, whereas those with a sample restricted to students of science lead teachers included 1,565 students and 59 schools.

In addition to all of the models just described, we present three supplementary analyses. The first examines whether the treatment affects students who are ELLs or schools with higher proportions of ELL students differently. The second investigates whether the treatment effect differs for schools with less experienced teachers. Finally, the third uses specific subsets of life science standards as outcome measures to test hypotheses about immersion unit and periodic test alignment.

Multilevel Model Estimates of the Treatment Effects

In our two-level models we added level-1 and level-2 predictors in steps, starting first with a model that included treatment status and the local district dummy codes to account for the randomization blocks. This model determines the treatment effect while controlling for the fixed effects of the randomization blocks. The next model added the school-level composite pretest measure of science achievements. This model should improve the power and precision of the coefficient for the impact of treatment assignment (Raudenbush, 1997). The third model added the student-level test timing dummy variable to account for variation in the test scores due to the timing of the instructional units and assessments. The timing of the instruction and tests could affect the observed outcomes primarily because of differences in the maturation level of students across the academic year and differences in students' level of overall exposure to science instruction. A second set of analytical models is restricted to scores of students of science lead teachers in the treatment and control schools. Because

Table 2. Multilevel models predicting student- and school-level life science achievement

	Schoolwide Sample				Science Lead Teacher Sample			
	Empty	Model 1	Model 2	Model 3	Empty	Model 1	Model 2	Model 3
Student-level timing dummy				–0.013				–0.029
				(0.028)				(0.033)
School-level intercept	0.558***	0.607***	0.220*	0.220*	0.566***	0.630***	0.145	0.140
	(0.013)	(0.039)	(0.089)	(0.087)	(0.022)	(0.055)	(0.158)	(0.153)
Treatment		–0.053*	–0.058**	–0.056**		–0.100*	–0.111**	–0.111**
		(0.023)	(0.020)	(0.019)		(0.042)	(0.039)	(0.038)
Science pretest composite			0.669***	0.678***			0.843**	0.883***
			(0.132)	(0.132)			(0.249)	(0.242)
District 1		0.055	–0.008	0.008		0.039	–0.028	–0.032
		(0.047)	(0.045)	(0.044)		(0.064)	(0.055)	(0.054)
District 2		–0.043	–0.048	–0.048		–0.039	–0.049	–0.058
		(0.048)	(0.047)	(0.045)		(0.102)	(0.105)	(0.104)
District 3		0.062	0.042	0.039		0.055	0.049	0.041
		(0.063)	(0.053)	(0.052)		(0.100)	(0.082)	(0.078)
District 4		–0.024	–0.029	–0.030		–0.016	–0.029	–0.035
		(0.049)	(0.047)	(0.046)		(0.072)	(0.071)	(0.069)
District 5		–0.052	–0.034	–0.034		–0.028	–0.012	–0.015
		(0.044)	(0.042)	(0.041)		(0.054)	(0.051)	(0.049)
District 6		–0.064	–0.029	–0.030		–0.040	0.014	0.015
		(0.040)	(0.037)	(0.036)		(0.062)	(0.059)	(0.057)
District 7		–0.084*	–0.025	–0.023		–0.097	–0.016	–0.002
		(0.040)	(0.040)	(0.039)		(0.051)	(0.054)	(0.054)

$^*p < .05$. $^{**}p < .01$. $^{***}p < .001$.

the science lead teacher at each of the study schools was determined prior to randomization, we are able to produce true experimental estimates of treatment assignment for this subset of teachers and students. These models follow the same sequence as just described. Models for the supplementary analyses are discussed in conjunction with their results.

Life Science Outcomes. In Table 2, we present the student- and school-level results for the life science outcome. The schoolwide results for all fourth-grade teachers and students are presented in the columns located on the left of the table, and the results for the restricted sample of fourth-grade science lead teachers and their students are found in the set of columns to the right of the table. The schoolwide two-level unconditional model suggests that there is statistically significant school-level variability for the outcome. We calculate the intraclass correlation using the formula

$$\rho = \frac{\tau}{\left(\tau + \sigma^2\right)},$$

where τ represents the variance between level-2 units, or schools, and σ^2 is the variance between level-1 units, or students. Using this formula results in an intraclass correlation of 0.25. This means that 25% of the total outcome variation lies between schools.

The first model for the schoolwide outcomes estimates the treatment effect controlling for the fixed effects of the eight local districts, which served as the randomization blocks. Under this model, there is a statistically significant, negative estimate of the treatment effect of –0.05, which translates into a 5 percentage point advantage for control schools on the life science outcome, on average. The second model estimates the treatment effect controlling for the randomization blocks and the aggregate score from the previous year's three science periodic scores. After statistically controlling for the aggregate pretest, the standard error of the treatment coefficient was reduced and the precision of the estimate was improved. The estimated impact suggested the scores of treatment schools were 0.06 points, or 6 percentage points, below those of control schools on average.

The third schoolwide model estimates the treatment effect, controlling for local district, school composite score on previous science assessments, and the student-level test timing dummy code indicating whether the life science assessment was administered in the first half of the school year. Taking into account the timing of the life science assessment did not affect the precision or magnitude of the estimated treatment effect. Thus, there was a reliable schoolwide difference between treatment and control schools on the fourth grade life science assessment, with control schools scoring 6 percentage points above treatment schools on average.

The second set of models in Table 2 shows the student- and school-level outcomes of each school's science lead teacher and his or her students. Again, the two-level unconditional model demonstrates that there is a statistically discernable group effect. In this case, the intraclass correlation is 0.52, indicating that fully 52% of the total variation for the life science outcome was between schools. (With only one classroom per school, this variance decomposition effectively combines school- and classroom-level variation at the school level.) The first model estimating the treatment effect after controlling for the randomization blocks revealed a statistically significant treatment effect. In this case, the estimated life science scores for students of science lead teachers from treatment schools was 0.10, or 10 percentage points, lower than those of students of science lead teachers from the control schools.

The second model for the science lead teacher sample estimates the treatment effect controlling for local district and aggregate score on the previous year's three science periodic scores. Under this model, the statistically significant negative treatment effect remained. Scores for treatment schools were 0.11 points, or 11 percentage points, below those of control schools on average. Finally, the third model estimates the treatment effect, controlling for local district, school composite score on previous science assessments, and the test timing dummy. Even after taking into account the timing of the life science assessment, the negative treatment effect remained statistically significant, indicating a control advantage of about 11 percentage points. The student-level test timing dummy was not a statistically significant predictor of the outcome. This suggested that the scores of the students who took the life science periodic in the first half of the academic year were statistically equivalent to those who took the assessment later during the academic year.

Earth and Physical Science Models. Results for the schoolwide and science lead teacher samples on the earth science periodic assessment outcome are presented in Table 3 and results for physical science are found in Table 4. These analyses follow the same progression as the modeling strategy used for the life science outcome. In both analyses of earth and physical science achievement, there were no statistically reliable differences between treatment and control. This result held for both the schoolwide and science lead teacher samples.

Supplementary Analyses for ELL Students

In addition to testing for the main effects of the treatment, we also tested whether there were differential effects of the treatment for ELL students or schools with large proportions of ELL students. We tested for these interaction effects because we were interested in whether science inquiry methods involving

Table 3. Multilevel models predicting student- and school-level earth science achievement

	Schoolwide Sample				Science Lead Teacher Sample			
	Empty	Model 1	Model 2	Model 3	Empty	Model 1	Model 2	Model 3
Student-level timing dummy				0.017				0.041
				(0.022)				(0.052)
School-level intercept	0.565***	0.551***	0.073	0.063	0.589***	0.556***	0.045	0.010
	(0.013)	(0.032)	(0.113)	(0.114)	(0.016)	(0.039)	(0.155)	(0.184)
Treatment		0.007	0.001	0.002		0.012	–0.000	0.003
		(0.022)	(0.017)	(0.017)		(0.030)	(0.024)	(0.026)
Science pretest composite			0.824***	0.830***			0.884**	0.919**
			(0.181)	(0.180)			(0.250)	(0.276)
District 1		0.128*	0.070	0.074		0.127*	0.065	0.076
		(0.048)	(0.043)	(0.042)		(0.054)	(0.045)	(0.043)
District 2		0.024	0.019	0.019		0.041	0.034	0.032
		(0.038)	(0.035)	(0.034)		(0.042)	(0.042)	(0.041)
District 3		0.029	0.003	0.001		0.055	0.046	0.043
		(0.086)	(0.073)	(0.071)		(0.110)	(0.094)	(0.089)
District 4		0.042	0.036	0.038		0.046	0.034	0.043
		(0.039)	(0.038)	(0.037)		(0.049)	(0.049)	(0.048)
District 5		−0.001	0.022	0.025		0.021	0.038	0.042
		(0.040)	(0.040)	(0.039)		(0.043)	(0.043)	(0.044)
District 6		−0.048	−0.006	−0.007		0.011	0.070	0.074
		(0.042)	(0.036)	(0.035)		(0.062)	(0.050)	(0.052)
District 7		−0.053	0.020	0.023		−0.077	0.017	0.023
		(0.037)	(0.037)	(0.037)		(0.051)	(0.054)	(0.058)

$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Table 4. Multilevel models predicting student- and school-level physical science achievement

	Schoolwide Sample				Science Lead Teacher Sample			
	Empty	Model 1	Model 2	Model 3	Empty	Model 1	Model 2	Model 3
Student-level timing dummy				–0.005				–0.039
				(0.034)				(0.023)
School-level intercept	0.632***	0.639***	0.275*	0.274*	0.663***	0.654***	0.213	0.220
	(0.013)	(0.036)	(0.113)	(0.114)	(0.017)	(0.050)	(0.166)	(0.167)
Treatment		–0.019	–0.017	–0.017		–0.001	–0.007	–0.006
		(0.024)	(0.022)	(0.022)		(0.031)	(0.028)	(0.027)
Science pretest composite			0.621**	0.621**			0.758**	0.755**
			(0.176)	(0.177)			(0.264)	(0.265)
District 1		0.090	0.047	0.047		0.048	–0.005	–0.000
		(0.050)	(0.042)	(0.043)		(0.062)	(0.051)	(0.050)
District 2		0.055	0.051	0.051		0.102	0.097	0.102
		(0.041)	(0.038)	(0.038)		(0.057)	(0.053)	(0.051)
District 3		–0.010	–0.026	–0.027		–0.026	–0.028	–0.025
		(0.069)	(0.062)	(0.062)		(0.096)	(0.094)	(0.093)
District 4		0.020	0.019	0.018		0.041	0.034	0.039
		(0.056)	(0.051)	(0.051)		(0.062)	(0.057)	(0.057)
District 5		–0.006	0.004	0.004		0.012	0.030	0.037
		(0.046)	(0.049)	(0.048)		(0.059)	(0.059)	(0.057)
District 6		–0.048	–0.019	–0.019		–0.036	0.014	0.015
		(0.042)	(0.043)	(0.043)		(0.068)	(0.067)	(0.065)
District 7		–0.071	–0.024	–0.024		–0.084	–0.014	–0.018
		(0.043)	(0.044)	(0.044)		(0.071)	(0.069)	(0.069)

$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

hands-on exploration could boost the science achievement of students whose first language was not English.

LAUSD has a large number of ELL students; according to their Web site (http://search.lausd.k12.ca.us/cgi-bin/fccgi.exe), in grades K-12 with approximately 38% of the students were classified as ELL in the 2006–07 school year. In the control group, 33% of the student body was ELL, compared to 32% of the treatment group. To test whether the treatment had differential effects on ELL students, we tested for a cross-level interaction effect using the following student-level model:

$$Y_{ij} = \beta_{0j} + \beta_{1j}(\text{TIMING})_{ij} + \beta_{2j}(\text{ELL})_{ij} r_{ij},$$

and the following level-2, or school-level, model

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{PRETEST})_j + \gamma_{02}(\text{TREATMENT})_j + \gamma_{03\ldots09}(\text{DISTRICT})_j + u_{0j},$$

$$\beta_{1j} = \gamma_{10},$$

$$\beta_{2j} = \gamma_{20} + \gamma_{21}(\text{TREATMENT}).$$

The results of this model indicated that ELL students in the schoolwide sample scored about 13 percentage points lower on average than non-ELL students on the life science periodic assessment but that this achievement gap did not differ across schools by their treatment status. In the science lead teacher sample, ELL students scored about 11 percentage points lower than non-ELL students, and again there were no within-school differences for this Effect × School-Level Treatment status. Therefore, we can conclude that our expectation that hands-on learning might have differential effects for ELL students is not supported by these data. The within-school gap between ELL and non-ELL students is substantial, but it is statistically equivalent within treatment and control schools.

Next, we tested whether the treatment effect varied according to a school's proportion of ELL students using the following student-level model

$$Y_{ij} = \beta_{0j} + \beta_{1j}(\text{TIMING})_{ij} + r_{ij},$$

and the following school-level model

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{PRETEST})_j + \gamma_{02}(\text{TREATMENT})_j + \gamma_{03\ldots09}(\text{DISTRICT})_j + \gamma_{10}(\%\text{ELL})_j$$
$$+ \gamma_{11}(\text{TREATMENT}^{*}\ \%\text{ELL})_j + u_{0j},$$

$$\beta_{1j} = \gamma_{10}.$$

According to the results for the schoolwide sample, the main effect of the school-level proportion of ELL students was statistically significant and negative, with a coefficient of –0.219 and a standard error of 0.102. However, the interaction between the treatment and school proportion of ELL students was not statistically significant, meaning that the treatment effect did not vary according to the school's percentage of ELL students. In the science lead teacher sample, neither the main effect of the school-level proportion of ELL students nor the interaction between the treatment and the proportion of ELL students were statistically significant. Based on these findings, we can conclude that the effect of the professional development in science inquiry did not vary according to the school's proportion of ELL students.

Supplementary Analyses on Potential Interactions of the Treatment with Teacher Experience

We also investigated whether the treatment had a different effect for schools with less experienced teachers. We might expect that the effect of being in the treatment group would differ according to a science teacher's teaching experience. For instance, one hypothesis may be that because less experienced teachers are less likely to have set lesson plans that they have used year after year, they may be more likely to adopt a new curriculum offered through the professional development. To test this hypothesis, we first created a dummy variable coded 1 if the teacher had taught 3 years or less. For the schoolwide sample, 18.2% of the teachers at control schools had taught 3 years or less, whereas 18.5% of the teachers at treatment schools had done so. For the science lead teacher sample, 12.5% of the science lead teachers in control schools had taught for 3 years or less, whereas in treatment schools 17% had done so. Neither of these differences in means was statistically significant. This variable was included at the school-level, meaning that for the schoolwide sample it represented the proportion of new teachers at the school, whereas for the science lead teacher sample it indicated whether the science lead teacher was new. We used the following student-level model to generate results

$$Y_{ij} = \beta_{0j} + \beta_{1j}(\text{TIMING})_{ij} + r_{ij},$$

and employed the following level-2 model

$$\begin{aligned}\beta_{0j} &= \gamma_{00} + \gamma_{01}(\text{PRETEST})_j + \gamma_{02}(\text{TREATMENT})_j + \gamma_{03\ldots09}(\text{DISTRICT})_j \\ &\quad + \gamma_{10}(\text{NEW TEACHER})_j + \gamma_{11}(\text{TREATMENT}^{*}\ \text{NEW TEACHER})_j + u_{0j}, \\ \beta_{1j} &= \gamma_{10}.\end{aligned}$$

In the schoolwide sample, neither the main nor interaction effect was statistically significant. For the science lead teacher sample, the main effect of being a new teacher was not statistically significant, but the interaction between the treatment and the new teacher dummy was statistically significant. The coefficient for the treatment indicator was –0.142 with a standard error of 0.041, whereas the coefficient for the interaction of the treatment and the new teacher dummy was 0.202 with a standard error of 0.082. This means that the treatment actually had a net positive effect, relative to the control, of about 6 percentage points for schools for science lead teachers who had 3 years of experience or less. Meanwhile, schools with more experienced science lead teachers scored about 14 percentage points lower than control schools. Thus, it appears that students of new teachers did benefit to a greater extent than students of more experienced teachers from their teachers' exposure to professional development in science inquiry.

Supplementary Analyses on Specific Standards

One possible explanation for the negative effect of being assigned to the treatment group on the life science score is that there is a lack of alignment between the immersion instruction and the periodic assessment. To address this concern, we conducted three supplementary analyses of subsets of standards covered on the periodic assessment. These analyses investigated three related hypotheses:

- According to teacher reports, many teachers had trouble making it all the way through the curriculum provided in the unit. Teachers may have covered the standards at the beginning of the unit in depth while neglecting to cover the rest. In this case, we might expect treatment students to do just as well as, or even better than, control students on test items measuring standards covered first in the unit.
- Although the immersion unit covers all California science standards for life science, some standards are more central to the inquiry activities. We might expect that students at treatment schools would score better than, or just as well as, control students on the standards that received emphasis in the unit.
- The science inquiry immersion unit utilized for the treatment condition contained scripted questions embedded in the unit as a method of formative assessment (REAP questions). REAP questions focused on recall and extension of new knowledge, ability to analyze and interpret new knowledge, ability to make predictions, and self-assessment of student learning. If teachers are assessing student knowledge using these formative assessments, we might expect students in the treatment group to score better on standards explicitly covered in the REAP questions.

The summary of findings from these analyses is available in Table 5.

First, we examined whether there was a difference in scores for standards covered early in the unit. We tested this hypothesis by seeing which standards were stressed in the first sections of the unit and using the combined score on these standards as an outcome measure in a multilevel model. There are six steps in the unit overall; we looked at the standards emphasized in the first two

Table 5. The immersion unit: Steps, lessons, and standards

Step	Lesson	Class Time	Key Concepts	Standards Covered	Standards Emphasized
Step 1	1.1	60 min	Producers require sunlight for growth and development.	2a, 3a, 3b, 6e	2a
	1.2	45 min	Producers convert sunlight into energy that can be used by living things.	2a, 2b, 6e	2a, 2b
Step 2	2.1	45 min	Organisms need matter to live and grow. Energy is transferred from the sun to producers and from producers to consumers.	2a, 2b	2a, 2b
	2.2	20 min	Consumers get energy by eating producers or other consumers.	2a, 2b	2a, 2b
Step 3	3.1	45 min	Ecosystems can be characterized by their living and nonliving components.	2b, 2c, 3a, 3d, 6a	2b, 2c, 6a
	3.2	60 min	Scientific progress is made by asking meaningful questions and conducting careful investigations.	6a, 6b, 6c, 6f	6a, 6c
	3.3	75 min	Scientists collect quantitative and qualitative data to support explanations.	6a, 6d, 6f	6a, 6f
	3.4	30 min	Scientists collect measurements and observations in order to formulate and justify explanations.	2c, 6a, 6b	2c, 6a, 6b

Step 4	4.1	30 min	Decomposers are consumers that recycle matter from plants and animals.	2b, 2c, 3d, 6a	2b, 2c, 3d, 6a
	4.2	60 min	A food web illustrates the transfer of energy among multiple food chains in an ecosystem.	2b, 3a	2b
Step 5	5.1	45 min	Scientists develop explanations using observations and what they already know about the world.	2a, 2b, 2c, 3a, 3b, 3c, 3d, 6a, 6c, 6e	2c, 6a, 6c, 6e
Step 6	6.1	20 min	In decomposition, dead matter is broken down and recycled by living organisms.	2c	2c
	6.2	60 min	Not all waste material can be recycled through the natural process of decomposition.	2c	2c

steps, both of which consist of two lessons. (Table 5 provides an overview of the steps, lessons, and standards covered in the unit.)

Using the break between Steps 2 and 3 of the unit as a cutoff point seemed reasonable because Step 3 has four lessons pertaining to an extended inquiry activity that teachers might not embark on if previous experiments ran over the allotted time. Because the first two steps emphasize Standards 2a, 2b, and 6a, we used the combined score on these standards as an outcome measure in our multilevel model. There were 10 test items that comprised this measure. The mean score on this outcome was 0.55, or 55% correct, with a range of 0 to 1 and a standard deviation of 0.24.

The multilevel models followed the same sequence as in the main analysis for the life science assessment. The schoolwide results for all fourth-grade students indicated that treatment schools, on average, scored about 6 percentage points lower than control schools on the standards covered first. This finding is virtually identical to that from the main analysis of the schoolwide sample as shown in Table 7.

For the sample of science lead teachers and their students, results from the fully specified model indicated that the treatment group scored about 9 percentage points lower than the control group, on average. Although the effect of being in the treatment group was still negative, the coefficients were slightly less negative than those in the models with the total life science score outcome. Treatment schools scored about 11 percentage points lower than control schools on average on the life science periodic assessment as a whole, while in this analysis they only scored lower by about 9 percentage points. This suggests that even though students of science lead teachers in the treatment schools were still scoring lower on the standards covered first in the unit, they were not scoring quite as poorly on these standards as on the assessment as a whole. Still, these results do not fully explain the negative treatment effect.

The second supplementary analysis examined how students scored on the standards emphasized throughout the immersion unit because of their centrality in inquiry activities. Although the unit covered all of the California science standards, Standards 2a, 2b, 2c, 3d, 6a, 6b, 6c, and 6e received special emphasis, as indicated by Table 6.

Although it would have been best to create an outcome measure consisting of a composite score on the aforementioned standards, the life science periodic assessment only included test items on 2a, 2b, 2c, 3d, 6a, and 6e; therefore, the composite score on these standards was the dependent variable. There were 16 test items that composed this outcome measure. The mean score on this outcome was 0.53, or 53% correct, with a range of 0 to 1 and a standard deviation of 0.22.

Again, these multilevel models, which are displayed in Table 7, followed the same sequence as in the main analysis of the life science total score. In the fully specified model for the schoolwide sample, treatment schools scored about 6 percentage points lower than control schools, on average. This result

Table 6. Alignment of California science standards for grade four

Standard for Life Science	Covered in Unit	Emphasized in Unit	Tested on Periodic Assessment for Life Science	No. of Items on Life Science Periodic Assessment[a]
2a	Yes	Yes	Yes	2
2b	Yes	Yes	Yes	7
2c	Yes	Yes	Yes	3
3a	Yes	No	Yes	3
3b	Yes	No	Yes	3
3c	Yes	No	Yes	2
3d	Yes	Yes	Yes	2
6a	Yes	Yes	No	1
6b	Yes	Yes	No	0
6c	Yes	Yes	No	0
6d	Yes	No	No	0
6e	Yes	Yes	Yes	1

[a]Total $N = 22$ items.

did not depart from the findings for the total score on the life science periodic assessment for the schoolwide sample. For the sample restricted to students of science lead teachers, there was a negative effect of about 10 percentage points of being in the treatment group. The negative effect of being in the treatment group was lower for the science periodic assessment as a whole than for these emphasized standards, but only by approximately 1 percentage point. This does not lend credence to the notion that treatment schools would do better on standards emphasized in the unit.

Finally, we tested the third hypothesis, that treatment students would perform just as well as or better than control students on standards emphasized in REAP questions, by creating an outcome measure consisting of scores on questions covering those standards. As the results in Table 7 suggest, the treatment estimates produced the treatment estimates produced by these models match the estimates obtained from the main analyses, meaning that this hypothesis holds no weight.

These supplementary analyses indicate that the imperfect alignment of subject matter topics in the immersion unit with those on the periodic assessment accounts for, at most, a small portion of the negative treatment effect. However, another sort of misalignment remains unexamined. Although the curriculum and assessment covered similar topics, they did not place the same sorts of cognitive demands on students. As Porter (2006) explained, achievement reflects the intersection of topic and cognitive demand more so than it reflects one or the other of these dimensions. Consequently, more careful consideration of the types of tasks that students carried out in the immersion unit in

Table 7. Comparison of estimated treatment effects for different subsets of standards

	Standards Subset Included in Outcome			
	Main Analysis All Standards	Standards Covered First 2a, 2b, 6a	Standards Emphasized in Unit 2a, 2b, 2c, 3d, 6a, 6e	Standards Targeted Through REAP 2a, 2b, 2c, 3a, 3b, 6a
Schoolwide treatment estimate	–0.056** (0.019)	–0.055* (0.021)	–0.057** (0.020)	–0.057** (0.020)
Science lead teacher treatment estimate	–0.111** (0.038)	–0.090* (0.037)	–0.098* (0.037)	–0.111** (0.038)

Note. The tabulated schoolwide and science lead teacher–only treatment effects were estimated after controlling for the covariates included in fully specified model.

comparison to those demanded on the periodic assessment (and other standardized tests) may further enlighten us about achievement differences between immersion and comparison schools.

DISCUSSION

The main research question for this article is, Does professional development for teachers in science immersion positively impact student science achievement? According to these 1st-year analyses, the treatment does have a statistically significant effect, though not in the expected direction. For the schoolwide sample, we see a negative effect of about one fourth of a standard deviation. The most proximal effects of the treatment should be found for the restricted science lead teacher sample on the life science outcome. Because the science lead teachers were the primary targets for participation in the summer professional development and because the professional development focused on the life science domain, we would expect impacts for this group and for this outcome to be strongest during the 1st year. Indeed, the effect was strongest for the science lead teacher sample on the life science outcome, but, as summarized in Table 8, it was a negative effect of half of one standard deviation. In contrast, the more distal effects of the treatment on the other two periodic assessment domains, earth and physical science, did not reveal treatment effects of any statistical or practical significance. Supplementary analyses of students taught by science lead teachers, the primary target of the teacher development opportunity, indicated that the intervention had positive effects for students of teachers with less than 3 years of experience, but larger, negative effects for students taught by veteran teachers.

How should one interpret these results and, most important, what may explain the negative 1st-year treatment impact on the targeted sample and outcome? The first explanation may be that the attrition rates were different and that perhaps more inclusive testing practices among the treatment schools caused the lower scores. For the schoolwide sample, this explanation does not seem to hold, as the treatment and control attrition rates because of

Table 8. Impacts of professional development on science achievement expressed as effect sizes

	Analytical Sample	
Assessment	Schoolwide Sample	Science Lead Teacher Sample
Life science	−0.27	−0.51
Earth science	0.01	−0.02
Physical science	−0.08	−0.03

missing data were statistically equivalent. In the sample of students of science lead teachers there was differential attrition between the treatment and the control groups, making it necessary to interpret those results with caution. However, given that we see a negative estimate of the treatment effect in both samples, this differential attrition may not hold that much sway over the estimates.

Another possibility might be that student maturation affected the outcome and because treatment students were more likely than control students to take the life science assessment before the earth or physical science assessments, this may have placed the treatment students and schools at a relative disadvantage on the life science test. However, given that the test timing dummy was not a statistically significant predictor of student scores, this explanation does not appear to hold weight.

Other explanations require greater exploration. For instance, it may be that implementing the new instructional and curricular approaches supported by the treatment was a complex task, which prevented teachers from covering all of the content assessed on the tests. Indeed, these complexities in implementing the nonroutine practice of immersion may have caused an implementation dip. That is, as Fullan and Miles (1992) described it, the stage of implementing a reform when things get worse before they get better. The implementation dip is literally a dip in performance and confidence as one encounters an innovation that requires new skills and new understandings. Working through the effects of the implementation dip requires time—often 2 to 3 years (Fullan, 1991). Although the current problem of the implementation dip in the context of this study is somewhat perplexing, it does seem to suggest that teachers and schools in LAUSD are, in fact, attempting to implement immersion. Although their initial attempts have resulted in some unexpected negative treatment effects, it is possible that these temporary dips in performance can be overcome. Support for this explanation comes from the finding that the negative effects were most pronounced among veteran teachers and that effects for novice teachers were positive. It is precisely the veteran teachers for whom the immersion curriculum represents a challenge to change their practice.

It would be hasty based on these 1st-year results to conclude that professional development for science immersion results will continue to result in declines in student achievement. When teachers depart from more traditional methods of teaching science, there is bound to be variation in success and fidelity to teaching methods learned in professional development. On a positive note, because there is a statistically significant result for life science but not earth or physical science, we can conclude that teachers are attempting to implement the immersion curriculum. Furthermore, qualitative evidence from focus groups conducted by LAUSD researchers indicates that teachers are excited about the science immersion units and eager to try implementing them again next year. Finally, the benefits for new teachers is an encouraging development. We, as researchers, look forward to seeing whether the treatment will have a positive effect in the future after teachers and schools have adjusted to the new science immersion teaching methods.

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REFERENCES

American Association for the Advancement of Science. (1989). *Science for all Americans*. Washington, DC: Author.

Boruch, R., May, H., Turner, H., Lavenberg, J., Petrosino, A., de Moya, D., et al. (2004). Estimating the effects of interventions that are deployed in many places: Place-randomized trials. *American Behavioral Scientist, 47,* 608–633.

California State Department of Education. (2004). *Key elements of testing*. Sacramento: Author.

Cornfield, J. (1978). Randomization by group: A formal analysis. *American Journal of Epidemiology, 108,* 100–102.

District and Local Profile: Los Angeles Unified School District Profile. LAUSDnet: The Official Website of the Los Angeles Unified School District.Retrieved June 25, 2008, from http://search.lausd.k12.ca.us/cgi-bin/fccgi.exe.

Donner, A., & Klar, N. (2000). *Design and analysis of group randomization trials in health research*. London: Arnold.

Fullan, M. (with Stiegelbauer, S.) (1991). *The new meaning of educational change*. New York: Teachers College Press.

Fullan, M., & Miles, M. (1992, September). Getting reform right: What works and what doesn't. *Phi Delta Kappan*, pp. 745–752.

Gamoran, A., Anderson, C. W., Quiroz, P. A., Secada, W. G., Williams, T., & Ashmann, S. (2003). *Transforming teaching in math and science: How schools and districts can support change*. New York: Teachers College Press.

Glennan, T. K., Jr., & Resnick, L. B. (2004). School districts as learning organizations: A strategy for scaling education reform. In T. K. Glennan, Jr., S. J. Bodilly, J. R. Gallagher, & K. A. Kerr (Eds.), *Expanding the reach of educational reforms: Perspectives from leaders in the scale-up of educational interventions* (pp. 517–563). Santa Monica, CA: RAND. Retrieved September 12, 2005, from http://www.rand.org/pubs/monographs/2004/RAND_MG248.pdf

Hoffer, R., & Cantrell, S. (2003). *Baseline report for the Los Angeles Urban Systemic Program in science, mathematics and technology* (Pub. No. 157). Los Angeles: LAUSD Planning, Assessment, and Research Division. Retrieved September 12, 2005, from http://notebook.lausd.net/pls/ptl/url/ITEM/DC60153E-2655EBA0E0330A081FB5 EBA0

Kelly, K. (2004). *Second year report on the Los Angeles Urban Systemic Program in mathematics, science, and technology* (Pub. No. 203). Los Angeles: LAUSD Planning, Assessment, and Research Division. Retrieved September 12, 2005, from http://notebook.lausd.net/pls/ptl/url/ITEM/E1CF4D043387F78CE0330A081FB5 F78C

Murray, D. M. (1998). *Design and analysis of group-randomized trials*. New York: Oxford University Press.

National Research Council. (1996). *National science education standards*. Washington, DC: National Academies Press.

Newmann, F. M., & Associates. (1996). *Authentic achievement: Restructuring schools for intellectual quality*. San Francisco: Jossey-Bass.

No Child Left Behind Act of 2001, Pub. L. No. 107-110, 115 Stat. 1425 (2002).

Olson, S., & Loucks-Horsley, S. (Eds.). (2000). *Inquiry and the National Science Education Standards: A guide for teaching and learning*. Washington, DC: National Academies Press.

O'Sullivan, C. Y., Lauko, M. A., Grigg, W. S., Qian, J., & Zhang, J. (2003). *The nation's report card: Science 2000*. Washington, DC: National Center for Education Statistics.

Porter, A. C. (2006). Curriculum assessment. In J. L. Green, G. Camilli, & P. B. Elmore (Eds.), *Handbook of complementary methods in education research* (pp. 141–159). Washington, DC: American Educational Research Association.

Raudenbush, S. W. (1997). Statistical analysis and optimal design for cluster randomized trials. *Psychological Methods, 2*, 173–185.

Romberg, T. A., Carpenter, T. P., & Dremock, F. (Eds.). (2005). *Understanding mathematics and science matters*. Mahwah, NJ: Erlbaum.

Schunn, C., Millar, T., Lauffer, D., & the SCALE Immersion Team. (2005). *Full-cycle science and design for all K–12 students: SCALE science immersion definitions and design*. Pittsburgh, PA: University of Pittsburgh, Learning Research and Development Center.

Scruggs, L. (2004). *Science assessments for grades 4–8* (Los Angeles Unified School District Reference Guide). Los Angeles, CA: Los Angeles Unified School District.

Wilson, M. R., & Berenthal, M. W. (2005). *Systems of state science assessment*. Washington, DC: National Academies Press.